

Principle

LCD1602

Generally, LCD1602 has parallel ports, that is, it would control several pins at the same time. LCD1602 can be categorized into eight-port and four-port connections. If the eight-port connection is used, then all the digital ports of the SunFounder Uno board are almost completely occupied. If you want to connect more sensors, there will be no ports available. Therefore, the four-port connection is used here for better application.

LCD1602 uses the standard 16-pin port, among which:

Pin 1 (GND): connected to Ground

Pin 2 (Vcc): connected to 5V power supply

Pin 3 (Vo): used to adjust the contrast of LCD1602; the level is lowest when it's connected to a positive power supply, and highest when connected to ground (you can connect a 10K potentiometer to adjust its contrast when using LCD1602)

Pin 4 (RS): register select pin, controlling where in the LCD's memory you are writing data to. You can select either the data register, which holds what goes on the screen, or an instruction register, where the LCD's controller looks for instructions on what to do next.

Pin 5 (R/W): to read/write signals; it reads signals when supplied with high level (1), and writes when low level (0) (in this experiment, you only need to write data to LCD1602, so just connect this pin to ground)

Pin 6 (E): An enable pin that, when low-level energy is supplied, causes the LCD module to execute relevant instructions

Pin 7 (D0-D7): pins that read and write data

A and K: controlling LCD backlight; K connects to GND, and A to 3.3V. Turn the backlight on and you can see the characters displayed clear in a dim